A Dual-Axis Resonant Accelerometer Based on Electrostatic Stiffness Modulation in Epi-Seal Process

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***Abstract*—We report on the implementation of a novel vacuum-encapsulated dual-axis resonant accelerometer utilizing electrostatic stiffness modulation. Novel flexure designs are used to enable fully-decoupled in-plane dual-axis translational motions of the proof-mass to achieve low cross-axis sensitivity with the translational mode frequencies above 20 kHz for high bandwidth applications. Differential frequency readout scheme mitigates the temperature effects on frequency stability of resonators and reduces off-axis sensitivity. The fabricated device measures a scale factor of 22Hz/g (119ppm/g) with low cross-axis sensitivity. The accelerometer demonstrates high performance with 18μg/√Hz VRW and 27μg BI in ambient environment. Optimization of the oscillator circuits and implementation of chip-level ovenization can further improve the performance of the design, with potentials to reach sub-μg performance for emerging high-bandwidth application.**

Keywords—dual-axis resonant accelerometer; wide bandwidth; micro-gravity accelerometer; electrostatic stiffness modulation

I. Introduction

The evolution of MEMS accelerometer technology has enabled a large variety of applications that improve the quality of people's daily lives [1-3]. Recently, the bloom of the IoT has created many emerging applications with a demand for lower noise and wider bandwidth accelerometers such as wearable microsystems for health informatics [4] and machine or structure health monitoring [5]. The most important metrics for such applications are small size, low noise and wide operational bandwidth, which allow the accelerometer to collect wide spectrum body sound information generated by internal organs in a convenient and noninvasive form. Therefore, high-bandwidth single-proof-mass multi-axis accelerometer designs that can deliver high performance with small form factors are desirable.

However, achieving such goals are challenging due to the tradeoffs between sensitivity, size and bandwidth in a MEMS accelerometer. For a conventional quasi-static capacitive accelerometer with amplitude readout, the performance is typically limited by the electrical noise in the interface circuit, interconnection parasitics and mismatches between differential capacitances [6]. In order to increase sensitivity and signal-to-noise-ratio, low-stiffness large-proof-mass designs are utilized, causing low bandwidth and large sensor size.

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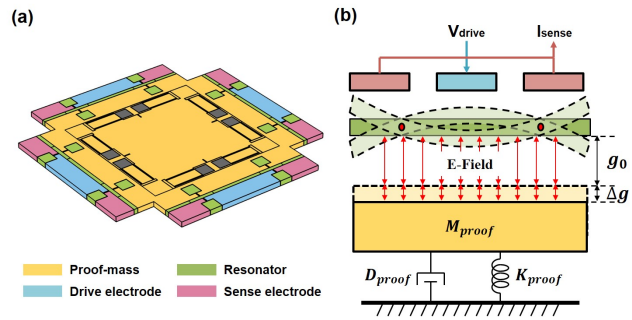

Figure 1. (a) Schematic of the bi-axial resonant accelerometer, (b) Schematic of gap-changing electrostatic spring softening effect (top view)

Alternatively, a frequency-output resonant accelerometer can be used to minimize the effects of paracitics and instrument noise. Earlier works on resonant accelerometers which utilize mechanical stiffness modulation have demonstrated good electrical noise rejection and high performance [7-11]. However, mechanical stiffness modulation designs require lever-structures to create large acceleration induced stress for high sensitivity, leading to increased sensor size. In addition, their dynamic range is usually limited by the mechanical stiffness nonlinearity of the low stiffness resonators used as the sensing elements.

Recently, resonant accelerometers based on electrostatic stiffness modulation have been demonstrated with the potential to achieve low noise, wide bandwidth and extended linear dynamic range through stiff structure and large electrostatic force [12,13]. A disadvantage of these designs is that they only have single-axis sensing capability, making multi degree-of-freedom sensing challenging. Therefore, developing wide bandwidth single proof-mass multi-axis accelerometer designs that can deliver high performance with small form factors are highly desirable.

Here, we present a dual-axis resonant accelerometer with electrostatically modulated frequency readout, for the first time, implemented using the Epi-Seal process [11], which simultaneously enables wide bandwidth and high resolution. The accelerometer utilizes four resonators for dual-axis differential frequency sensing and a unique two degrees of freedom flexure mechanism which provides mechanically decoupled proof-mass motions along the two sensing axis. Detailed design and characterizations of the accelerometer are presented in the following sections.

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II. DUAL AXIS RESONANT ACCELEROMETER

Fig. 1(a) shows the schematic of dual axis in-plane resonant accelerometer. The design consists of four capacitively transduced in-plane free-free beam resonators surrounding a proof-mass within a 2mm×2mm footprint. The resonators and proof-mass are biased at different DC voltages during operation, which introduces electrostatic spring softening effect to the resonators through the parallel-plate capacitive gaps between the resonators and the proof-mass. Differential resonator pairs are placed on opposite ends of the proof-mass. When external acceleration is applied, the gap for one resonator becomes smaller and the gap for the other resonator along the same sensing axis becomes larger (Fig.1(b)). Which causes the amount of electrostatic stiffness to change and modulate the resonant frequencies of the two differential resonators.

The differential frequency readout scheme provides several benefits. First, the identical resonators have similar frequencies and TCFs. Therefore, frequency fluctuations due to temperature variations can be largely canceled with the differential measurement. Secondly, it also suppresses the first order nonlinear effects in gap-closing stiffness modulation mechanism. Furthermore, the tuning area changes induced by accelerations orthogonal to the sensing axis create same effects on the two resonators, so the frequency difference remains unchanged.

For a single-proof-mass dual-axis design, minimizing cross-axis sensitivity is critical. In the proposed design, this is ensured by the flexure design combining doubly folded beam and single beam with mechanically decoupled deformations under accelerations in different directions (Fig.2).

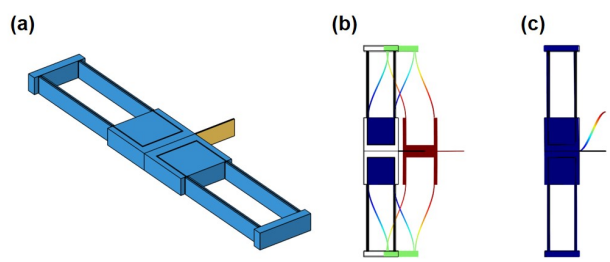

Figure 2. Two degrees of freedom flexural structure of the accelerometer. (a) Doubly folded flexure (blue), and beam flexure (yellow). (b) Deformation in x-axis. (c) Deformation in y-axis.

For example, for the flexure with the orientation shown in Fig. 2(b) and 2(c), the doubly folded beam provides lower stiffness in x-axis and extremely high stiffness in y-axis, which allows the single beam flexure to move along with the proof-mass in x direction with negligible deformation. Similarly, the single beam has lower stiffness in y-axis as shown in Fig. 2(c), allowing proof-mass to move freely in y direction without causing significant deformations in the doubly folded beam structure.

The designed resonant characteristics of proof-mass is shown in Fig.3. The in-plane translational modes have resonant frequency of 21.2kHz, which will provide wide bandwidth near

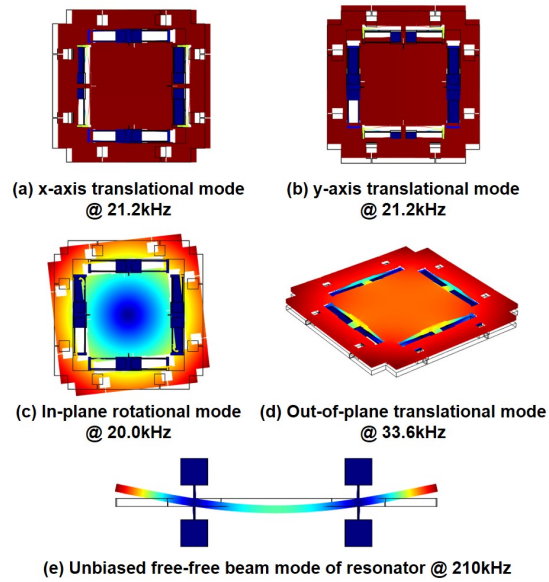

Figure 3. (a-d) Resonant characteristics of two degrees of freedom proof-mass and (e) free-free beam resonator

20kHz with proper damping design. The in-plane rotational mode and out-of-plane translational mode also have high resonant frequencies at 20.0kHz and 33.6kHz, and frequency shift from such undesired motion can be canceled with the differential readout.

The thick free-free beam resonator design provides high translational stiffness which prevents large movement of the resonator due to external acceleration and electrostatic tuning force to reduce pull-in instability with high DC bias voltages. The free-free beam mode is the lowest mode which enables high frequency tunability without spurious modes near the resonant frequency.

III. EXPERIMENTAL RESULTS

The device was fabricated and wafer-level packaged using DYVI Epi-Seal process with 2mm×2mm die size. The high vacuum packaging of current Epi-seal process results in an

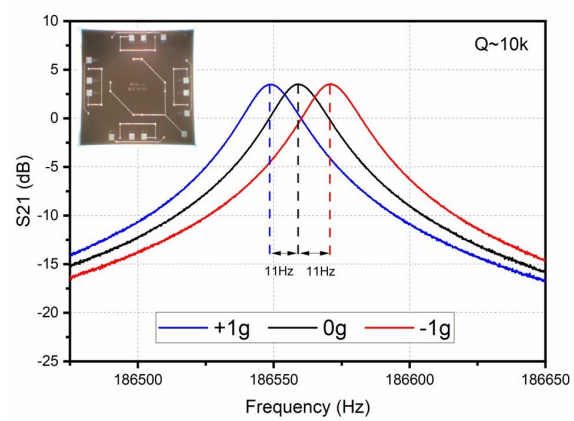

Figure 4. Measured resonant characteristics of a single resonator with different orientation

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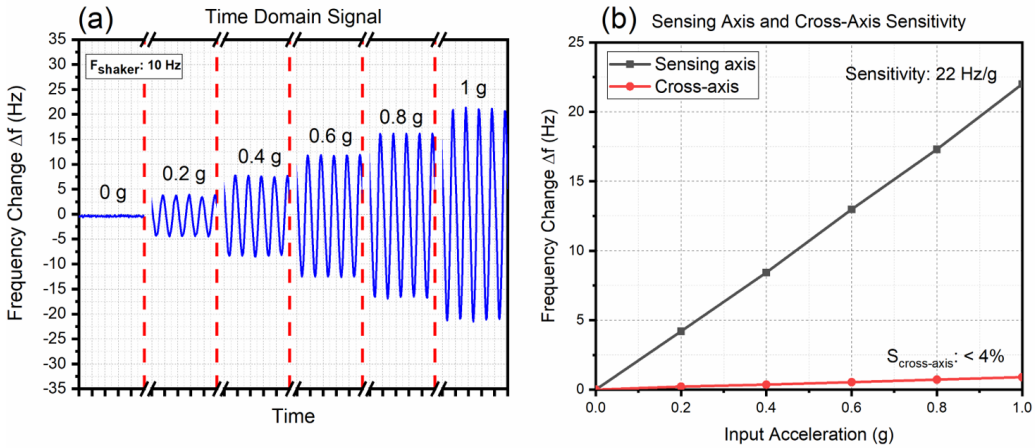

Figure 5 Scale factor measurement result. (a) Response to AC accelerations up to ±1g applied by a shaker table. (b) Scale factor measures 22Hz/g with cross axis sensitivity less than 4%.

under-damped condition for the proof-mass. Modifications to the process or future designs with added damping electrodes can create a near critically-damped condition to enable stable accelerometer operation over the full bandwidth of ~20kHz. The fabricated device was mounted and wirebonded to a PCB with TIAs built by discrete electronics and interfaced with an HF2LI lock-in amplifier with PLL for oscillator implementation. The resonators of the device showed resonant frequencies of ~188kHz and Q ~10k with 35V bias voltage and 0.5mV drive voltage (Fig.4).

The scale factor measurement was done by applying up to ±1g acceleration at 10Hz using a shaker table. The measurements show ~22Hz/g linear scale factor for both axis with a bias voltage of 35V,and a cross-axis sensitivity less than 4%, limited by the manual alignment accuracy between device and fixture for the skater table (Fig.5). It should be noted that the simulated cross axis sensitivity of the design is near zero.

The noise performance of the accelerometer was measured in ambient environment without temperature control. The long-term frequency fluctuations were observed on both resonators. The fluctuations mainly come from the ambient temperature variations and possibly environmental vibrations, which showed near-identical trend for the differential pair of resonators (Fig.6). The differential frequency output of the same measurement period shows that large fluctuations are cancelled leaving only small linear drift, possibly due to TCF mismatch between the resonators. The ADEV measurement of the differential frequency output showed a 27µg BI and a 18 µg/√Hz VRW (Fig. 7).

IV. Conclusion

We introduced and demonstrated a wideband dual-axis resonant accelerometer based on electrostatic stiffness modulation effect, fabricated and encapsulated in vacuum using the epi-seal process . A combined flexural structure design was implemented to provide good decoupling between x and y-axis motion. The experimental results showed scale factor of 22Hz/g with low cross-axis sensitivity, VRW of 18µg/√Hz, and BI of 27µg in ambient environment. Improved frequency stability is expected with on-chip ovenization, promising sub-µg BI.

Acknowledgment

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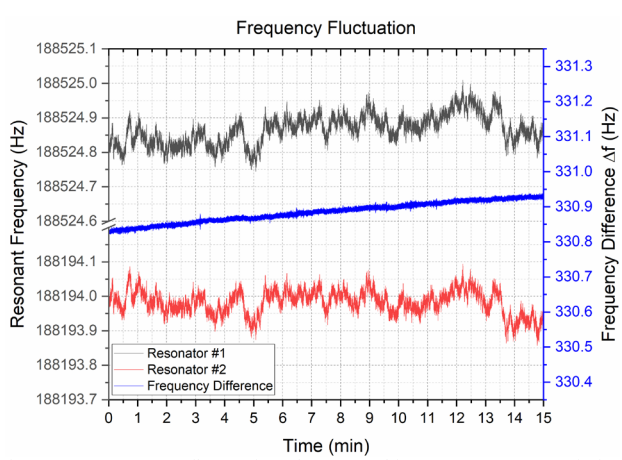

Figure 6. Frequency fluctuation due to ambient temperature variation. The frequency difference shows 6.5mHz/1min of drift.

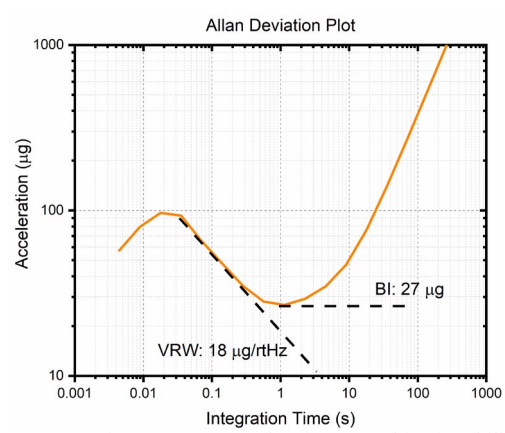

Figure 7. ADEV plot measurement showing 27µg bias instability at 1s integration time and 18µg/√Hz velocity random walk

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